Università della Svizzera italiana

Faculty of Informatics

Bioinformatics

# Homework #4

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# 1 Inactive state

We start by searching the canonical full-length A2AR sequence from UniProt, and then we copy and paste the sequence inside colab to predict the structure of A2AR monomer using AlphaFold2. As asked in the assignment, we run the code two times:

- The first one for a Single-sequence prediction (with no template)
- The second one with a full Multiple Sequence Alignment search (with no template)

First of all we define the metrics used by AlfaFold2 to describe how good the prediction is. The first one is the plDDT that say how well the predicted structure matches the true local structure of the protein with a value that is in $[0, 100]$, so that higher the value, higher the accuracy. We consider an high value of confidence a plDDT value greaten then 90 (color blue), i.e. both backbone and side chain prediction are reliable; and we consider instead a low value of confidence a plDDT value lower than 50 (color red). The second metric used is the PAE metrics or Predicted Aligned Error, that is measured in Angrostroms and tells us if relative position of different domains are actually correct. A low value of PAE tells us that low predicted error, while an high value means that AF2 is not confident in the relative position of two part of the protein. Is usually represented witha heat map, that is usually green/blue if the value of PAE is low and white if the value is high.

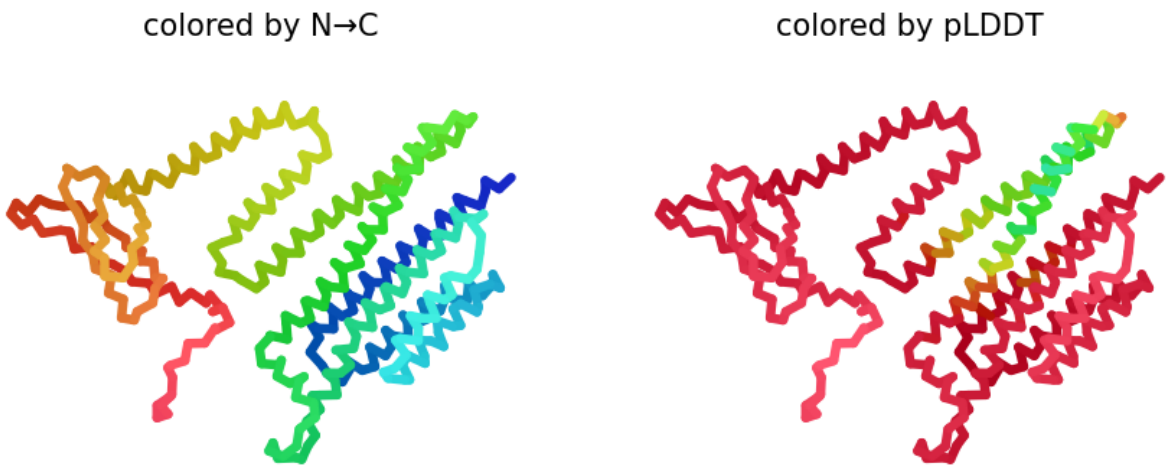


Figure 1

We can see in the 1 the first case (Single-sequence prediction). From the image on the right that the per-residue measure of local confidence for each section is low since we have almost all the prediction that is red, so that there is no confidence in the predicted structure. This analysis match with the value of the pLDDT, that is equal to 46.9 at the end. Note that this is the best one in the 5 model that the method have generated.

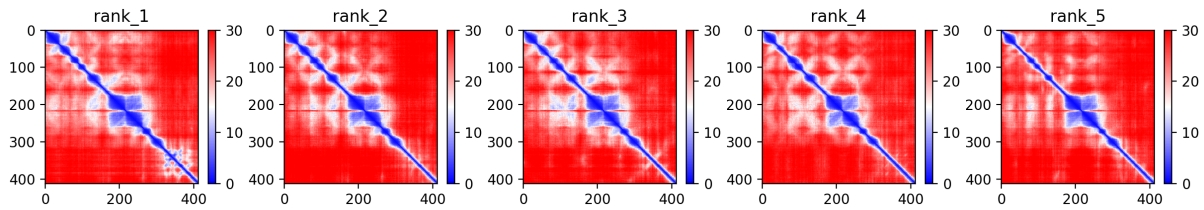


Figure 2

The 2 show us instead in the five heat map above the PAE values, confirming the AF2 is not confident in the relative position of almost all the two possible combination part of the protein, indeed only the main diagonal is blue (have a low PAE value) since only the nearest residues are correctly positioned.
Finally, we note in the 3 (a screen of the 3D structure) that the TM6 is the only one (along with TM5) to have a local pLDDT with a decent value (even with only a value of $\approx 70$) so that the prediction seems to have well-structured TM6 as an $\alpha$-helix, but is not able to gives the complete and compact structure of the protein.

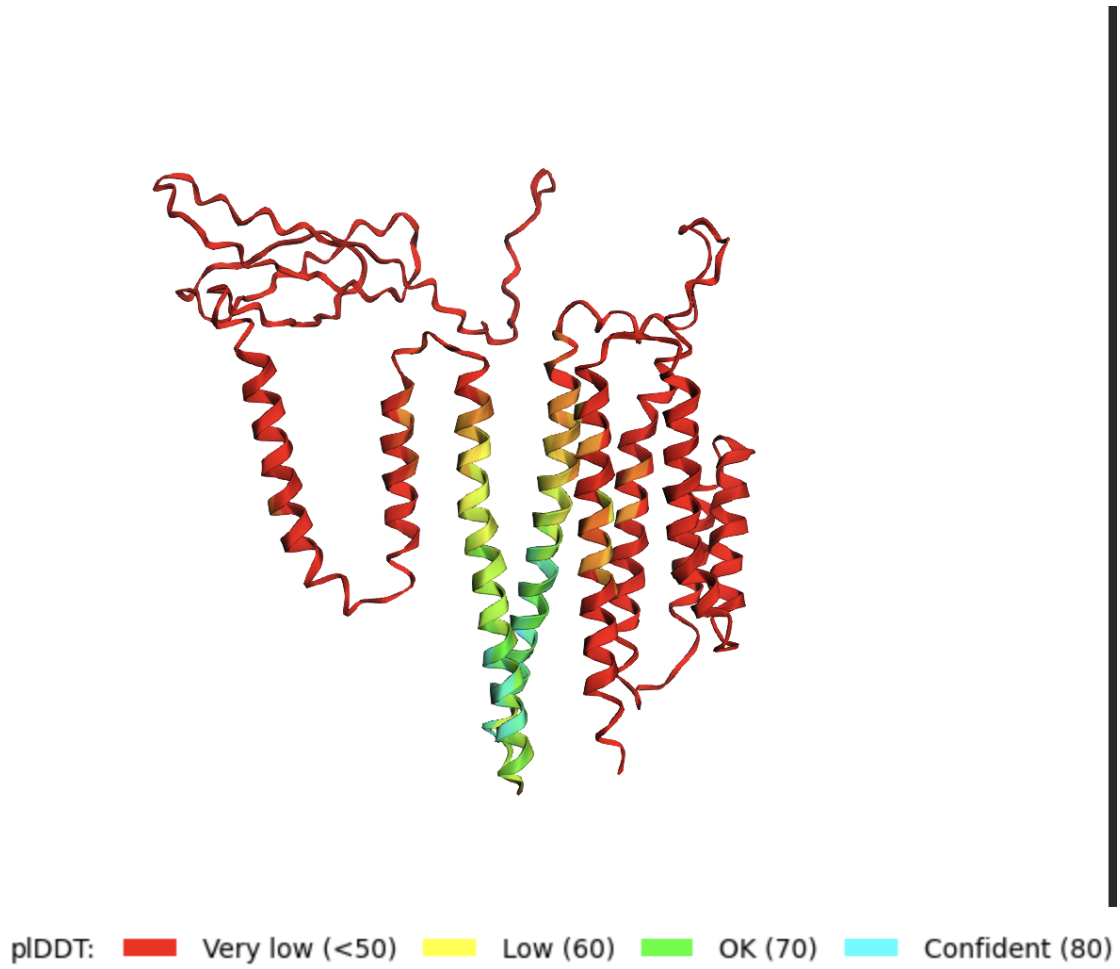


Figure 3

To confirm that using the single sequence prediction, the method fails to create the structure of the protein we can align the predicted structure to the canonical 3PWH (the ocher

one) using the PDB pairwise structure alignment to see that the canonical 3PWH does not overlaps with the prediction.

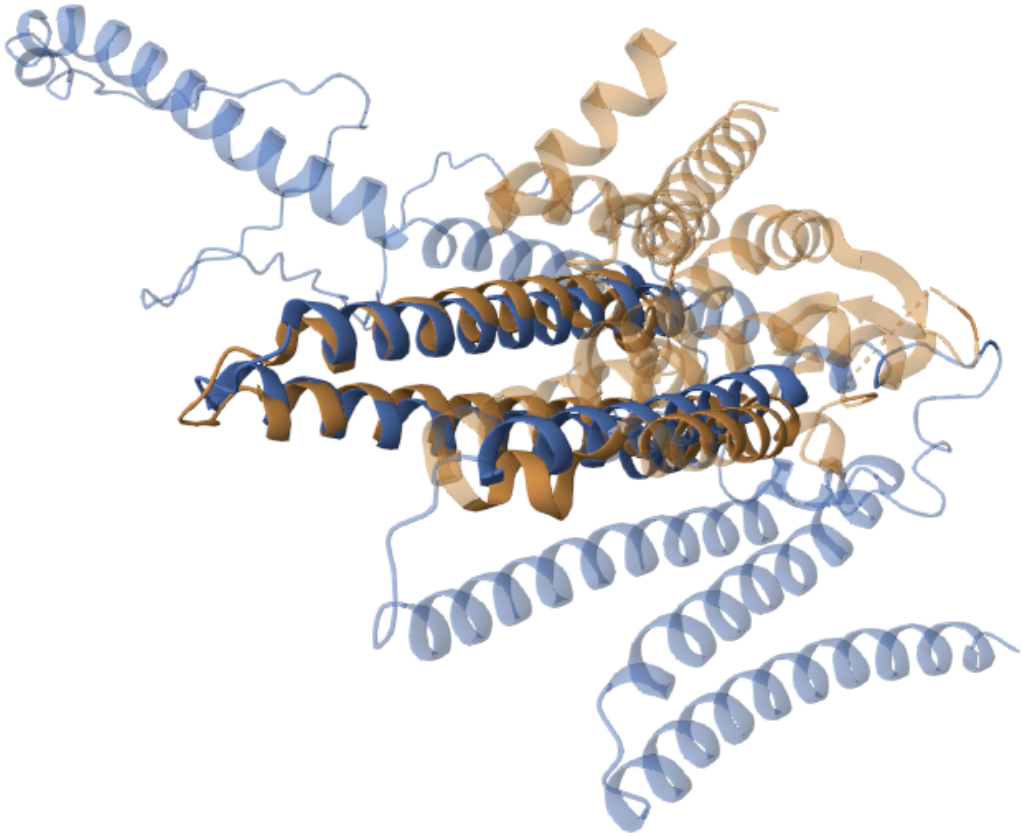

Figure 4

We know show the result of the run with full MSA. As we can see in 5, we know have a more compact result and looking at the right image, we note that almost all the helix have a blue color, denoting a plDDT value almost perfect ($> 90$), i.e. the structure of the protein is almost prefectly predicted. However, the best model out of the five evaluated gives a global plDDT value of only 75.4, this because we can see a long tail in the terminal part completely wrong, that decrease the mean value of plDDT.

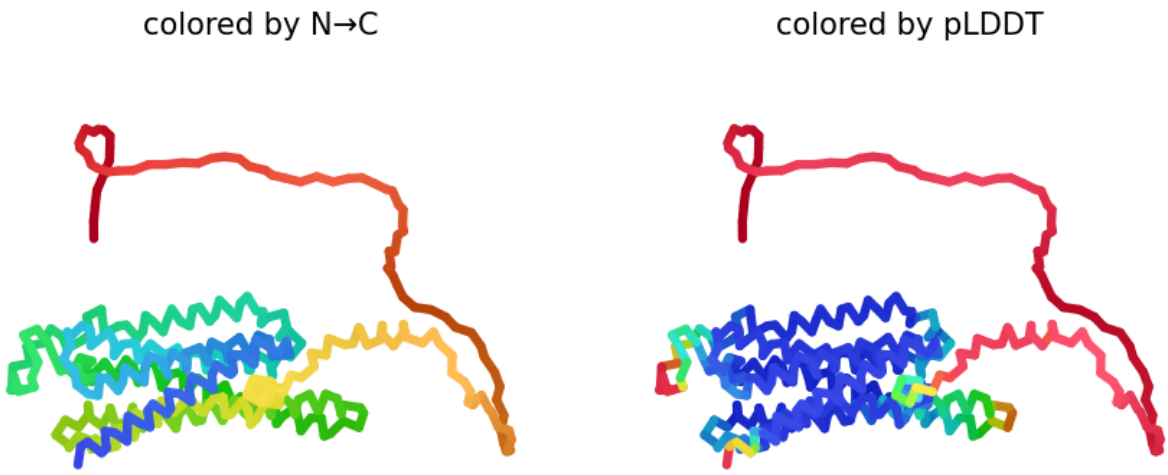


Figure 5

Looking the heat map in the 6 we confirm what we have said before, indeed we know have a first part (the square on the left-up side, that is almost everywhere blue) with a lower value of PAE, i.e. there is low error in the predicted relative position between two residual. As said before, the tail of the protein is completely wrong, indeed, the other part of the plot have an high value of PAE (red color).

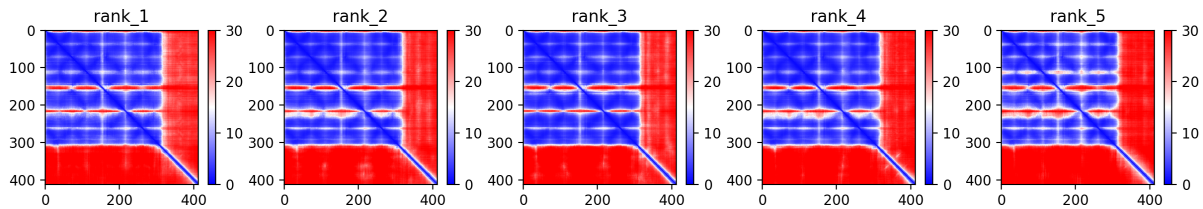


Figure 6

In contrast to the previous run, we can see in 7 the 3D structure of the protein with now, the TM6 well positioned inside (its the one green/yellow in the left image of 5 and the one the connect with the fifth in the 7 with a small loop, the one colored with yellow) and with the other TransMembrane helix compose the compact structure of the protein.

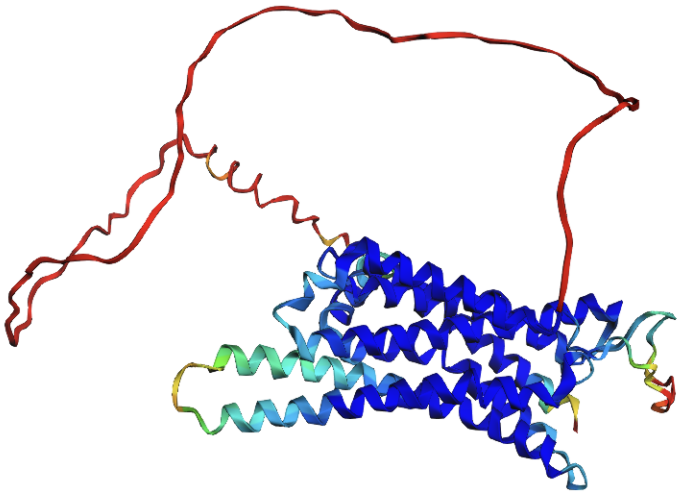

Figure 7

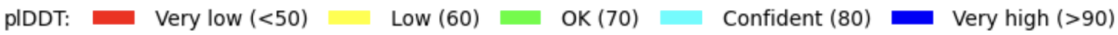


Figure 8

We finally confirm the things said before showing the pairwise structure alignment on PDB to see that the prediction overlaps almost everywhere the canonical 3PWH with all the 7 TransMembrane helix; with the exception of the long tail, that is the wrong part of the prediction (as already mentioned above).

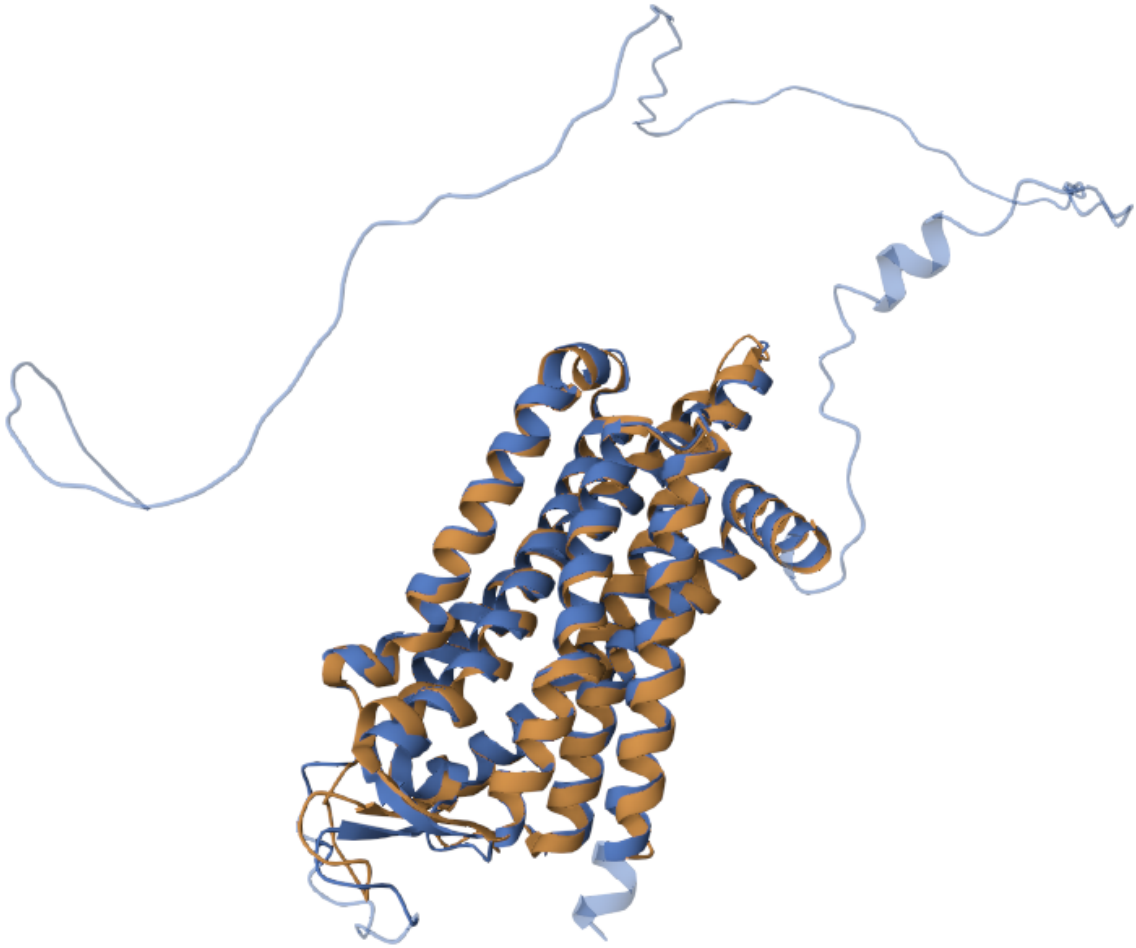

Figure 9

So we can finally say that the prediction with MSA more closely resembles the 3PWH inactive state, this because the MSA use the other residual to try to find the distance between the different TransMembrane to build the compact structure of the protein, seeing how these other residual evolved and trying to understand the structure of the protein by seeing if different part evolved simultaneously. On the other hand, the Single-sequence prediction try to predict the structure of the protein only looking at the FASTA sequence. We can conclude saying that the AF2, push the receptor toward inactive geometry (indeed we can see that in both the prediction the TM6 is closed) since is more stable.

## 2 Activation by mini-G Protein Coupling

To study the active state, we look for PDB 5G53, where is used a mini-G to stabilizes the active conformation.
The use of a G-protein in this case, actually allow us to move to an active state of the protein, where we can see that the TM6 is now open so mini-G contact the intracellular face near TM5/TM6 (we can see that in the 10, with an arrow that indicate the TM6 that is now open).

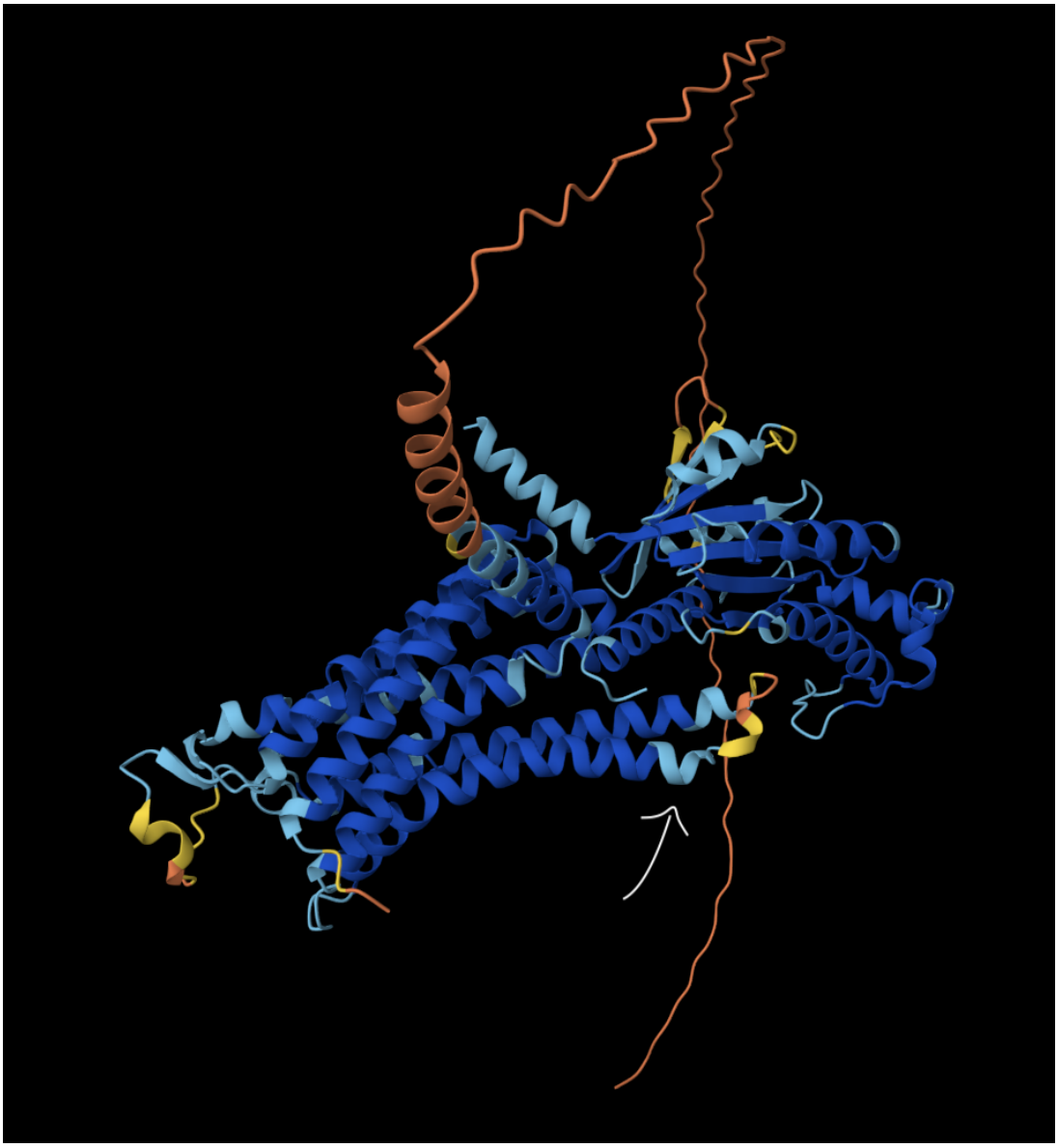

Figure 10

In the figure 11 we can see the portion of mini-G that contact intracellular face, with a residue of mini-G selected

Figure 11

We can also compare this to the inactive state that we have seen before, i.e. 3PWH. Indeed we clearly see in the 12 that in the case of 5G53 (the one color ocher) the TM6 is slightly curved, so that the G-protein can enter, while in the 3PWH case (the blue one) the TM6 is closed and continues straight, exactly where in 5G53 we have now the G-protein.

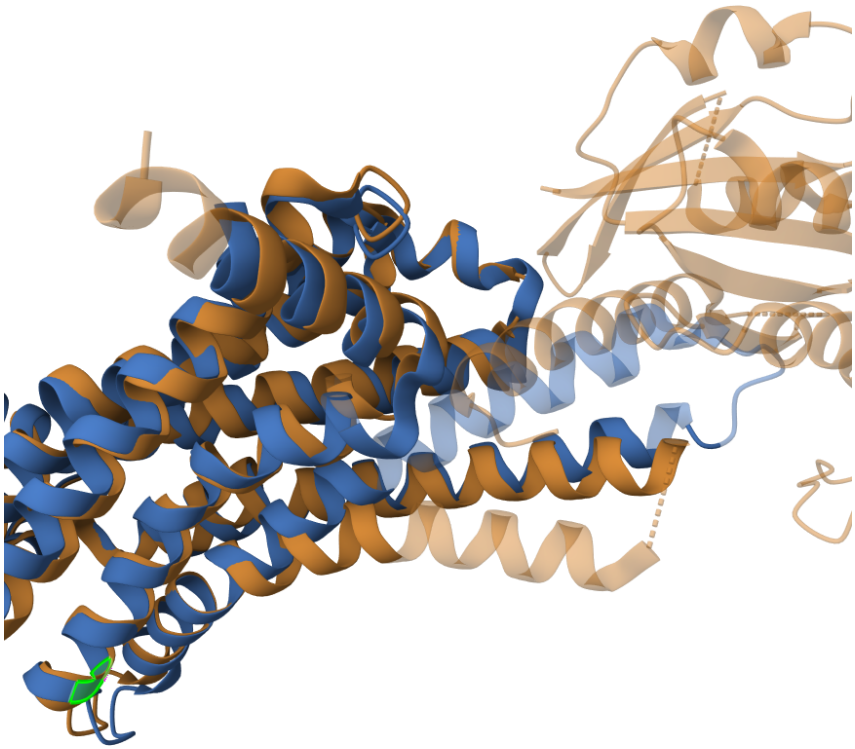

Figure 12

The ipTM score of 0.84 shows an high confidence and also the the PAE heat map gives us a good result, with most of the heat map green (small value); so that the AF3 is confident that the G-protein binds in that specific position.

So we conclude saying that, AF2 and AF3 have some architectural differences, of which the most important is the fact that in AF3 we have a diffusion module, where we start with a cloud of random Gaussian noise and learns to 'denoise' these positions to reveal the final structure. Another key difference is that in AF2, information was mixed directly across different rows of the MSA. Instead in AF3, all the information as to flow via the pair representation, because the pair representation must contain as much information as possible since is the backbone of the prediction. Unlike AF2, which was trained primarily on protein chains, AF3 can predict the structure of DNA and RNA and also can model the ligand, that instead in AF2 are ignored. The most important limitations are the fact that both AF2 and AF3 predict a static structure, and do not directly show what happens when, for example, TM6 is opening; the fact that struggles with "orphan" proteins (those with few close relatives) and finally the tendency to occasionally produce overlapping ("clashing") atoms in the predictions